

# Geodesic Newton Direction with Blended RHS

## 1 Setup

Let  $b_0$  denote the original affine term,  $c$  the original cost,  $e$  the identity element,  $A$  the constraint operator, and  $P(X)$  the quadratic representation  $P(X)v = XvX$ . The Gram matrix is

$$G = A^T P(W) A.$$

## 2 Three-term decomposition

The Newton step at the current scaling  $W$  is decomposed into three components that are independent of any blending parameters.

### 2.1 Gram solves

$$\begin{aligned} G y_0 &= A^T (2W), \\ G y_1^{(0)} &= -(c + A^T P(W) b_0), \\ G y_1^{(\theta)} &= c + A^T P(W) b_0 - A^T (e + P(W) e). \end{aligned}$$

### 2.2 Direction components

$$\begin{aligned} d_0 &= e - P(W^{1/2})(A y_0), \\ d_1^{(0)} &= P(W^{1/2})(-b_0 - A y_1^{(0)}), \\ d_1^{(\theta)} &= P(W^{1/2})(b_0 - e - A y_1^{(\theta)}). \end{aligned}$$

All six quantities  $(y_0, y_1^{(0)}, y_1^{(\theta)}, d_0, d_1^{(0)}, d_1^{(\theta)})$  depend only on  $W$ , the original data  $(b_0, c)$ , and the constraint operator  $A$ . They are computed with one factorization and three back-solves.

### 3 Parameterized direction

Given independent parameters  $\tau$  (original-problem weight) and  $\theta$  (identity-centering weight), the Newton direction is

$$d(k, \tau, \theta) = d_0 + k(\tau d_1^{(0)} + \theta d_1^{(\theta)}),$$

which is affine in  $k$ ,  $\tau$ , and  $\theta$ .

The corresponding dual variable is

$$y(k, \tau, \theta) = y_0 + k(\tau y_1^{(0)} + \theta y_1^{(\theta)}).$$

#### 3.1 Special cases

- $\tau = 1, \theta = 0$ : original problem,  $d = d_0 + k d_1^{(0)}$ .
- $\tau = 0, \theta = 1$ : feasibility/centering problem with  $b = e, c = A^T e$ ,  $d = d_0 + k d_1^{(\theta)}$ .
- $\tau = 1 - \theta$ : single-parameter blend  $b_\theta = \theta e + (1 - \theta) b_0, c_\theta = \theta A^T e + (1 - \theta) c$ .

### 4 Minimum-norm $k$ at fixed $(\tau, \theta)$

Define  $\tilde{d}_1 = \tau d_1^{(0)} + \theta d_1^{(\theta)}$ . The squared norm of the direction is

$$\|d(k)\|^2 = \|d_0\|^2 + 2k \langle d_0, \tilde{d}_1 \rangle + k^2 \|\tilde{d}_1\|^2,$$

which is minimized at

$$k^*(\tau, \theta) = -\frac{\langle d_0, \tau d_1^{(0)} + \theta d_1^{(\theta)} \rangle}{\|\tau d_1^{(0)} + \theta d_1^{(\theta)}\|^2}.$$

### 5 Slack, primal variable, and multiplier

The Newton direction  $d$  lives in  $v$ -space, where  $W = \exp(v)$ . The linearized primal slack and dual multiplier are recovered from  $d$  as follows.

#### 5.1 Slack

The primal slack (cone element) is

$$s = \frac{1}{k} P(W^{-1/2})(e - d).$$

## 5.2 Dual multiplier

The dual multiplier is

$$\lambda = \frac{1}{k} P(W^{1/2})(e + d) = \frac{1}{k} W \circ (e + d).$$

At the central path ( $d = 0$ ),  $s = \lambda = W/k$  and  $s \circ \lambda = W^2/k^2 = \mu W^2$ , where  $\mu = 1/k^2$ .

## 5.3 Primal variable

The optimization variable is

$$x = \frac{y}{k} = \frac{1}{k} y_0 + \tau y_1^{(0)} + \theta y_1^{(\theta)}.$$

## 5.4 Complementarity

The complementarity gap is

$$\langle s, \lambda \rangle = \frac{1}{k^2} \langle P(W^{1/2})(e - d), P(W^{1/2})(e + d) \rangle = \mu (m - \|d\|^2),$$

where  $m$  is the rank (dimension of the cone) and  $\mu = 1/k^2$ . At the central path ( $d = 0$ ), the gap is  $\mu m$ .

## 6 Duality residual

The Newton direction satisfies the identity

$$b_0^T \lambda + c^T x + \frac{\mu}{\tau} = \theta (\langle b_0, e \rangle + 1),$$

where  $\lambda$ ,  $x$ ,  $\mu = 1/k^2$  are as defined above. This can be verified by substituting the expressions for  $\lambda$  and  $x$  in terms of the decomposition.

At the central path ( $d = 0$ ,  $\tau = 1$ ,  $\theta = 0$ ), the left-hand side reduces to  $b_0^T \lambda + c^T x + \mu$ , and the right-hand side vanishes, recovering the standard complementarity relation.

At the trivially centered point ( $\theta = 1$ ,  $\tau = 1$ ,  $k = 1$ ,  $W = e$ ,  $d = 0$ ):  $\lambda = e$ ,  $x = 0$ ,  $\mu = 1$ , and

$$b_0^T e + 0 + 1 = \langle b_0, e \rangle + 1,$$

which is satisfied identically.

The *duality residual* is the absolute error in this equation:

$$r_{\text{dual}} = \left| b_0^T \lambda + c^T x + \frac{\mu}{\tau} - \theta (\langle b_0, e \rangle + 1) \right|.$$

## 7 Joint $(k, \tau)$ selection with $\theta = \mu = 1/k^2$

Setting  $\theta = \mu = 1/k^2$  ties the centering weight to the barrier parameter. The direction becomes

$$d = d_0 + k\tau d_1^{(0)} + \frac{1}{k} d_1^{(\theta)},$$

which depends on two free parameters  $(k, \tau)$ .

### 7.1 Inner-product basis

All norms and inner products of  $d$  can be expressed in terms of six scalars computed once from the decomposition:

$$\begin{aligned} a &= \|d_0\|^2, & b &= \langle d_0, d_1^{(0)} \rangle, & c &= \langle d_0, d_1^{(\theta)} \rangle, \\ p &= \|d_1^{(0)}\|^2, & q &= \langle d_1^{(0)}, d_1^{(\theta)} \rangle, & r &= \|d_1^{(\theta)}\|^2. \end{aligned}$$

### 7.2 Optimal $\tau$ at fixed $k$

The squared norm

$$\|d\|^2 = a + 2k\tau b + \frac{2c}{k} + k^2\tau^2 p + 2\tau q + \frac{r}{k^2}$$

is quadratic in  $\tau$ . Setting  $\partial\|d\|^2/\partial\tau = 0$ :

$$\tau^*(k) = -\frac{k b + q}{k^2 p}.$$

### 7.3 Reduced objective in $k$ alone

Substituting  $\tau^*(k)$  back:

$$\|d\|^2(k, \tau^*(k)) = a + \frac{2c}{k} + \frac{r}{k^2} - \frac{(k b + q)^2}{k^2 p}.$$

This is a rational function of  $k$  involving only the six precomputed scalars. It is minimized by a one-dimensional search over  $k > 0$ .

### 7.4 Algorithm (per factorization)

1. Compute the decomposition  $(d_0, d_1^{(0)}, d_1^{(\theta)})$ : one factorization and three back-solves.
2. Compute the six inner products  $a, b, c, p, q, r$ .
3. Line-search over  $k > 0$  to minimize  $\|d\|^2(k, \tau^*(k))$ .
4. Set  $\tau = \tau^*(k)$ ,  $\theta = 1/k^2$ ,  $\mu = 1/k^2$ .

5. Evaluate  $d = d_0 + k\tau d_1^{(0)} + (1/k) d_1^{(\theta)}$ .
6. Take the geodesic step  $W \leftarrow W \cdot \exp(\alpha d)$  with  $\alpha = \min(1, 2/\|d\|_\infty^2)$ .

No additional Gram solves are required beyond the initial decomposition. The entire parameter selection is determined by inner products of the precomputed direction components.

## 8 Derivation of the decomposition

Starting from the single-parameter blend  $b_\theta = \theta e + (1 - \theta) b_0$ ,  $c_\theta = \theta A^T e + (1 - \theta) c$ , the combined RHS for  $y_1$  is

$$\begin{aligned} G y_1 &= -(c_\theta + A^T P(W) b_\theta) \\ &= -(c + A^T P(W) b_0) + \theta (c + A^T P(W) b_0 - A^T e - A^T P(W) e), \end{aligned}$$

giving  $y_1 = y_1^{(0)} + \theta y_1^{(\theta)}$ . Expanding  $d_1 = P(W^{1/2})(-b_\theta - A y_1)$  yields  $d_1 = d_1^{(0)} + \theta d_1^{(\theta)}$ , which corresponds to  $\tau = 1 - \theta$ . Decoupling  $\tau$  and  $\theta$  generalizes this to the full two-parameter family.